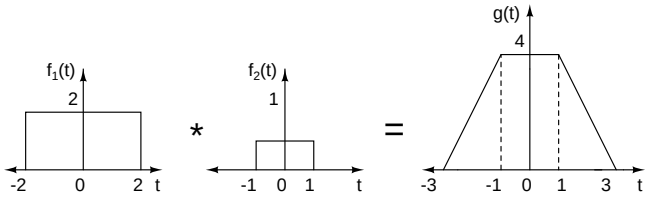
EE1101 : Signals and Systems

Tutorial 8 Solutions

1. (a) Given $f_1(t) = 2\,\text{rect}\left(\frac{t}{4}\right)$ and $f_2(t) = \text{rect}\left(\frac{t}{2}\right)$



$$g(t) = f_1(t) * f_2(t) = \begin{cases} 2(t+3) & , -3 \le t \le -1 \\ 4 & , -1 < t \le 1 \\ 2(3-t) & , 1 < t \le 3 \\ 0 & , |t| \ge 3 \end{cases}$$

(b) Using the analysis equation,

$$\begin{aligned} G(j\omega) &= \int_{-\infty}^{\infty} g(t)e^{-j\omega t}dt \\ &= \int_{-3}^{-1} 2(t+3)e^{-j\omega t}dt + \int_{-1}^{1} 4e^{-j\omega t}dt \\ &\quad + \int_{1}^{3} 2(3-t)e^{-j\omega t}dt \end{aligned}$$

Using the formulae $\int xe^{cx} = \left(\frac{cx-1}{c^2}\right)e^{cx}$ and $\int e^{cx} = \frac{1}{c}e^{cx}$,

$$\begin{aligned} G(j\omega) &= \left[\frac{(-2j\omega t - 2 - 6j\omega)e^{-j\omega t}}{-\omega^2}\right]_{-3}^{-1} + \left[\frac{4e^{-j\omega t}}{-j\omega}\right]_{-1}^{1} \\ &\quad + \left[\frac{(2j\omega t + 2 - 6j\omega)e^{-j\omega t}}{-\omega^2}\right]_{1}^{3} \\ &= \frac{(2+4j\omega)e^{j\omega} - 2e^{j3\omega}}{\omega^2} + \frac{8\sin\omega}{\omega} \\ &\quad + \frac{(2-4j\omega)e^{-j\omega} - 2e^{-j3\omega}}{\omega^2} \\ &= \frac{4\cos(\omega) - 8\omega\sin(\omega) - 4\cos(3\omega)}{\omega^2} + \frac{8\sin\omega}{\omega} \\ &= \frac{4[\cos(\omega) - \cos(3\omega)]}{\omega^2} \\ &= \frac{8\sin(2\omega)\sin(\omega)}{\omega^2} \\ &= 16\,\text{sinc}(2\omega/\pi)\,\text{sinc}(\omega/\pi) \end{aligned}$$

where $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$.

(c) We know that

$$\begin{aligned} \text{rect}\left(\frac{t}{\tau}\right) &\longleftrightarrow \tau\,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right) \\ f_1(t) = 2\,\text{rect}\left(\frac{t}{4}\right) &\longleftrightarrow 2(4)\,\text{sinc}(2\omega/\pi) \\ f_2(t) = \text{rect}\left(\frac{t}{2}\right) &\longleftrightarrow 2\,\text{sinc}(\omega/\pi) \end{aligned}$$

Using the convolution property of Fourier Transform,

$$\begin{aligned} G(j\omega) &= F_1(j\omega)F_2(j\omega) \\ &= 8\,\text{sinc}(2\omega/\pi)2\,\text{sinc}(\omega/\pi) \\ &= 16\,\text{sinc}(2\omega/\pi)\,\text{sinc}(\omega/\pi) \end{aligned}$$

(d) The magnitude spectrum of $G(j\omega)$ is plotted using matlab in Figure 1. Phase spectrum is plotted in Figure 2.

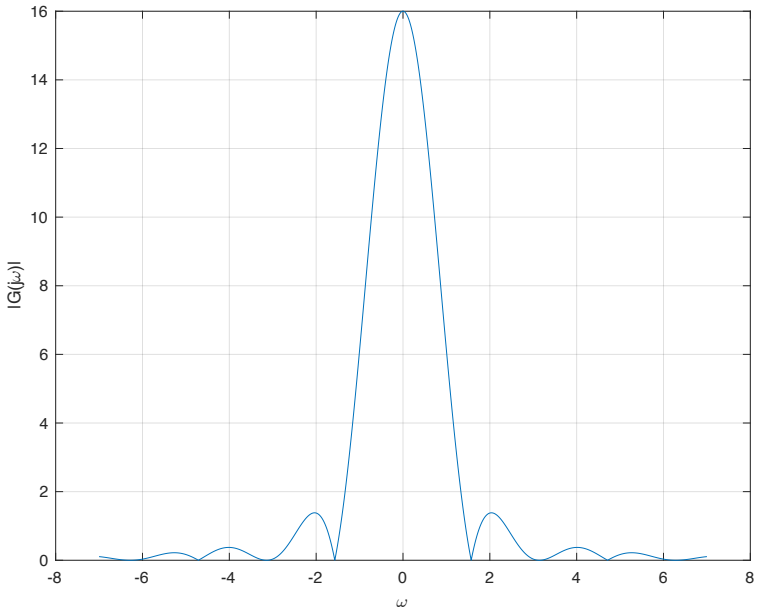


Figure 1: Magnitude spectrum of $G(j\omega)$ in Q2 generated using matlab

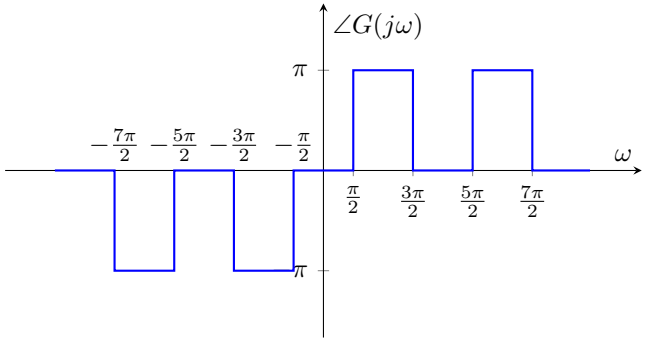
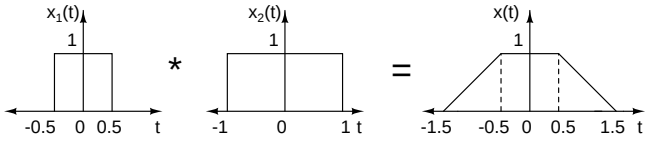


Figure 2: Phase spectrum of $G(j\omega)$ in Q2

Note: $\angle G(j\omega)$ can take either π or $-\pi$ for values of ω for which $G(j\omega)$ is negative. Here we have taken π for when ω is positive and $-\pi$ for when ω is negative to imply the point that phase spectrum is an odd function for real-valued time-domain signals.

2. The given signal $x(t)$ can be written as,

$$x(t) = x_1(t) * x_2(t)$$
$$\text{where, } x_1(t) = \text{rect}(t) \text{ and } x_2(t) = \text{rect}\left(\frac{t}{2}\right)$$



Using the convolution property of Fourier Transform,

$$\begin{aligned} X(j\omega) &= X_1(j\omega)X_2(j\omega) \\ &= \text{sinc}\left(\frac{\omega}{2\pi}\right) 2\,\text{sinc}\left(\frac{\omega}{\pi}\right) \\ &= 2\,\text{sinc}\left(\frac{\omega}{\pi}\right)\text{sinc}\left(\frac{\omega}{2\pi}\right) \end{aligned}$$

3. (a) Given $X(j\omega) = \dfrac{2\sin(3(\omega - 2\pi))}{(\omega - 2\pi)}$

$$\text{rect}\left(\frac{t}{\tau}\right) \longleftrightarrow \tau\,\text{sinc}\left(\frac{\omega\tau}{2\pi}\right) = \frac{2}{\omega}\sin\left(\frac{\omega\tau}{2}\right) \qquad (1)$$

Comparing with standard form in (1),

$$\text{rect}\left(\frac{t}{6}\right) \longleftrightarrow \frac{2}{\omega}\sin(3\omega)$$

Using the frequency shifting property of Fourier Transform,

$$e^{j2\pi t}\,\text{rect}\left(\frac{t}{6}\right) \longleftrightarrow \frac{2\sin(3(\omega - 2\pi))}{(\omega - 2\pi)}$$

(b) Given $X(j\omega) = cos(4\omega + \frac{\pi}{3})$

$$\begin{aligned} X(j\omega) &= cos(4\omega + \frac{\pi}{3}) \\ &= \frac{1}{2}\left[e^{j(4\omega+\frac{\pi}{3})} + e^{-j(4\omega+\frac{\pi}{3})}\right] \\ &= \left(\frac{e^{j\frac{\pi}{3}}}{2}\right)e^{j4\omega} + \left(\frac{e^{-j\frac{\pi}{3}}}{2}\right)e^{-j4\omega} \end{aligned}$$

We know that,

$$\delta(t) \longleftrightarrow 1$$

By using time shifting property,

$$\delta(t+4) \longleftrightarrow e^{j4\omega}$$
$$\delta(t-4) \longleftrightarrow e^{-j4\omega}$$

By using linearity property,
$x(t) = \left(\dfrac{e^{j\frac{\pi}{3}}}{2}\right)\delta(t+4) + \left(\dfrac{e^{-j\frac{\pi}{3}}}{2}\right)\delta(t-4)$

4. (a)

$$\begin{aligned} X(j\omega) &= \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\mathrm{d}t \\ &= \int_{-\infty}^{\infty} e^{-a|t|}e^{-j\omega t}\mathrm{d}t \\ &= \int_{-\infty}^{0} e^{at}e^{-j\omega t}\mathrm{d}t + \int_{0}^{\infty} e^{-at}e^{-j\omega t}\mathrm{d}t \\ &= \left[\frac{e^{(a-j\omega)t}}{a-j\omega}\right]_{-\infty}^{0} + \left[\frac{e^{(-a-j\omega)t}}{-a-j\omega}\right]_{0}^{\infty} \\ &= \frac{1}{a-j\omega} + \frac{1}{a+j\omega} \\ &= \frac{2a}{a^2+\omega^2} \end{aligned}$$

(b) From the duality property of Fourier transform, we know that

$$\begin{aligned} &\text{if} \quad x(t) \longleftrightarrow X(\omega) \\ &\text{then} \quad X(t) \longleftrightarrow 2\pi \cdot x(-\omega). \end{aligned}$$

Thus from part (a), we have

$$\begin{aligned} &\frac{2a}{a^2+t^2} \longleftrightarrow 2\pi \cdot e^{-a|\omega|} \\ \implies &\frac{1}{1+t^2} \longleftrightarrow \pi \cdot e^{-|\omega|} \end{aligned}$$

Using Parseval's theorem we have,

$$\begin{aligned} \int_{-\infty}^{\infty}\left[\frac{1}{1+t^2}\right]^2\mathrm{d}t &= \frac{1}{2\pi}\int_{-\infty}^{\infty}\left[\pi \cdot e^{-|\omega|}\right]^2\mathrm{d}\omega \\ &= \frac{\pi}{2}\int_{-\infty}^{0} e^{2\omega}\mathrm{d}\omega + \frac{\pi}{2}\int_{0}^{\infty} e^{-2\omega}\mathrm{d}\omega \\ &= \frac{\pi}{2}\left[\frac{e^{2\omega}}{2}\right]_{-\infty}^{0} + \frac{\pi}{2}\left[\frac{e^{-2\omega}}{-2}\right]_{0}^{\infty} \\ &= \frac{\pi}{2} \end{aligned}$$

5. a)

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\mathrm{d}t$$
$$X(0) = \int_{-\infty}^{\infty} x(t)e^{-j(0)t}\mathrm{d}t$$
$$X(0) = \int_{-\infty}^{\infty} x(t)\mathrm{d}t$$
$$X(0) = \int_{-1}^{0} 1\mathrm{d}t + \int_{0}^{1}(-t+1)\mathrm{d}t + \int_{1}^{2}(t-1)\mathrm{d}t + \int_{2}^{3} 1\mathrm{d}t$$
$$\implies X(0) = 3$$

b)

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\mathrm{d}\omega$$

$$x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega(0)}\mathrm{d}\omega$$

$$x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\mathrm{d}\omega$$

$$\int_{-\infty}^{\infty} X(j\omega)\mathrm{d}\omega = 2\pi x(0)$$

$$\implies \int_{-\infty}^{\infty} X(j\omega)\mathrm{d}\omega = 2\pi$$

c)

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}\mathrm{d}\omega$$

$$x(1) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega(1)}\mathrm{d}\omega$$

$$x(1) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega}\mathrm{d}\omega$$

$$\int_{-\infty}^{\infty} X(j\omega)e^{j\omega}\mathrm{d}\omega = 2\pi x(1)$$

$$\implies \int_{-\infty}^{\infty} X(j\omega)e^{j\omega}\mathrm{d}\omega = 2\pi.0$$

$$\implies \int_{-\infty}^{\infty} X(j\omega)e^{j\omega}\mathrm{d}\omega = 0$$

d)

Using Parseval's theorem,

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2\mathrm{d}\omega = \int_{-\infty}^{\infty} |x(t)|^2\mathrm{d}t$$

$$\implies \int_{-\infty}^{\infty} |X(j\omega)|^2\mathrm{d}\omega = 2\pi\left[\int_{-1}^{0} 1^2\mathrm{d}t + \int_{0}^{1}(1-t)^2\mathrm{d}t + \int_{1}^{2}(t-1)^2\mathrm{d}t + \int_{2}^{3} 1^2\mathrm{d}t\right]$$

$$= 2\pi\left[1 + \frac{1}{3} + \frac{1}{3} + 1\right] = \frac{16\pi}{3}$$

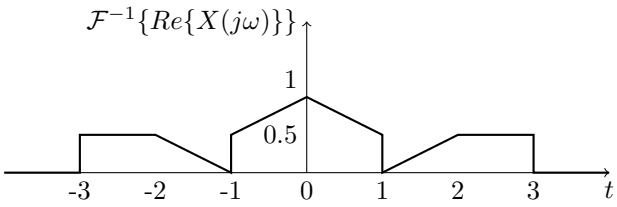
e)

$$Re\{X(j\omega)\} = \frac{X(j\omega) + X^*(j\omega)}{2}$$

$$\mathcal{F}^{-1}\{Re\{X(j\omega)\}\} = \frac{\mathcal{F}^{-1}\{X(j\omega)\} + \mathcal{F}^{-1}\{X^*(j\omega)\}}{2}$$

$$= \frac{x(t) + x^*(-t)}{2}$$

$$= \frac{x(t) + x(-t)}{2} \longrightarrow Ev\{x(t)\}$$



6. For the given signal, $T_0 = 1, \omega_0 = 2\pi$.

(a) Fourier series coefficients P_n of function $p(t)$:

$$P_n = \frac{1}{T_0}\int_{\frac{-1}{4}}^{\frac{1}{4}} p(t)e^{-jn\omega_0 t}dt, \quad n \neq 0$$

$$= \int_{\frac{-1}{4}}^{0}(1+4t)e^{-j2\pi nt}dt + \int_{0}^{\frac{1}{4}}(1-4t)e^{-j2\pi nt}dt$$

$$= \frac{4\sin^2(\pi n/4)}{\pi^2 n^2}, \quad n \neq 0.$$

$$P_0 = \int_{\frac{-1}{4}}^{\frac{1}{4}} p(t)dt = \frac{1}{4}.$$

Note: This can be solved using convolution property as well. Convolution of rect function with itself will give triangular function. In Fourier domain that manifests as $\mathrm{sinc}^2()$ function.

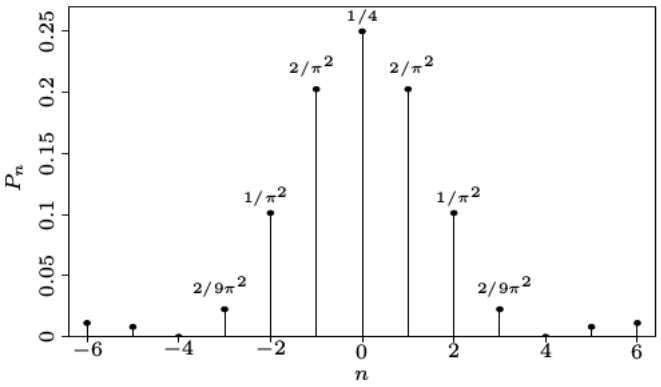


Figure 3: Sketch for P_n vs. n in Q6(a)

(b) Fourier transform of function $p(t)$:

By using Fourier series expansion,

$$p(t) = \sum_{n=-\infty}^{\infty} P_n e^{j2\pi nt}$$

$$P(j\omega) = \sum_{n=-\infty}^{\infty} \mathcal{F}\{P_n e^{j2\pi nt}\}$$

$$= \sum_{n=-\infty}^{\infty} P_n \mathcal{F}\{e^{j2\pi nt}\}$$

$$= \sum_{n=-\infty}^{\infty} 2\pi P_n \delta(\omega - 2\pi n)$$

$$= \sum_{n=-\infty}^{\infty} \frac{8\sin^2(\pi n/4)}{\pi n^2}\delta(\omega - 2\pi n)$$

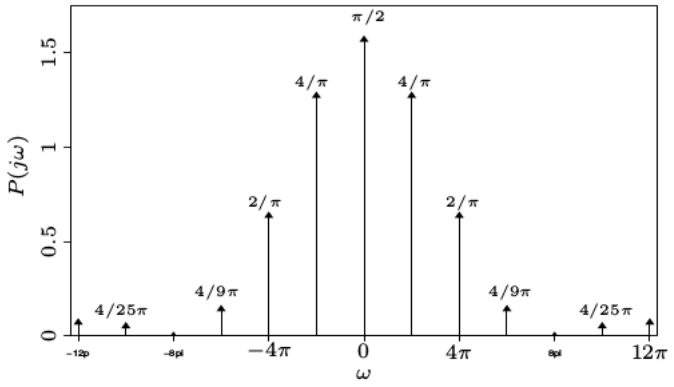


Figure 4: Sketch for $P(j\omega)$ vs. ω in Q6(a)

(c) $y(t) = p(t) \cdot x(t)$. Then the Fourier transform of $y(t)$ is

$$\begin{aligned} Y(j\omega) &= \frac{1}{2\pi}\{P(j\omega) * X(j\omega)\} \\ &= \frac{1}{2\pi}\Big\{2\pi \sum_{k=-\infty}^{\infty} P_k \delta(\omega - 2\pi k) * X(j\omega)\Big\} \\ &= \sum_{k=-\infty}^{\infty} P_k X(j(\omega - 2\pi k)) \\ &= \sum_{k=-\infty}^{\infty} \frac{4\sin^2(\pi k/4)}{\pi^2 k^2} X(j(\omega - 2\pi k)) \end{aligned}$$

(d) $x(t) = \text{sinc}(t)$, has the Fourier transform

$$X(j\omega) = \text{rect}\left(\frac{\omega}{2\pi}\right).$$

$$\begin{aligned} Y(j\omega) &= \frac{1}{2\pi}\{P(j\omega) * X(j\omega)\} \\ &= \sum_{k=-\infty}^{\infty} \frac{4\sin^2(\pi k/4)}{\pi^2 k^2} \text{rect}\left(\frac{\omega - 2\pi k}{2\pi}\right). \end{aligned}$$

7. (a)

$$\begin{aligned} X(j\omega) &= \int_0^{\infty} e^{-at} e^{-j\omega t} \mathrm{d}t \\ &= \left[\frac{e^{(-a-j\omega)t}}{-a-j\omega}\right]_0^{\infty} \\ &= \frac{1}{a+j\omega} \end{aligned}$$

Therefore,

$$\begin{aligned} Y(j\omega) &= H(j\omega)X(j\omega) \\ &= \frac{1}{(a+j\omega)^2} \end{aligned}$$

(b)

$$\frac{1}{(a+j\omega)^2} = j \cdot \frac{\mathrm{d}}{\mathrm{d}\omega}\left(\frac{-1}{a+j\omega}\right)$$

Using frequency differentiation property, we have

$$\begin{aligned} j \cdot \frac{\mathrm{d}}{\mathrm{d}\omega}\left(\frac{1}{a+j\omega}\right) &\longleftrightarrow t \cdot e^{-at}u(t) \\ \implies \frac{1}{(a+j\omega)^2} &\longleftrightarrow t \cdot e^{-at}u(t) \end{aligned}$$

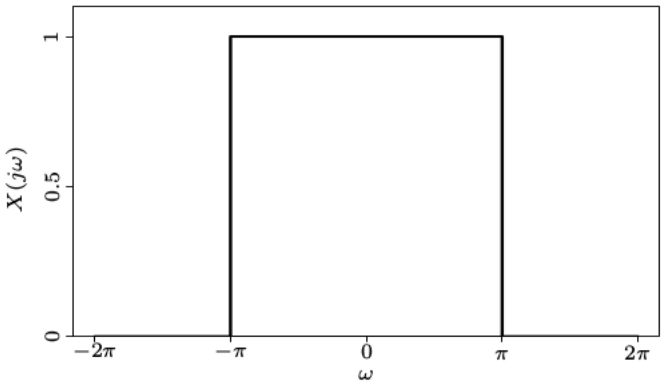


Figure 5: Sketch for $X(j\omega)$ in Q6(c)

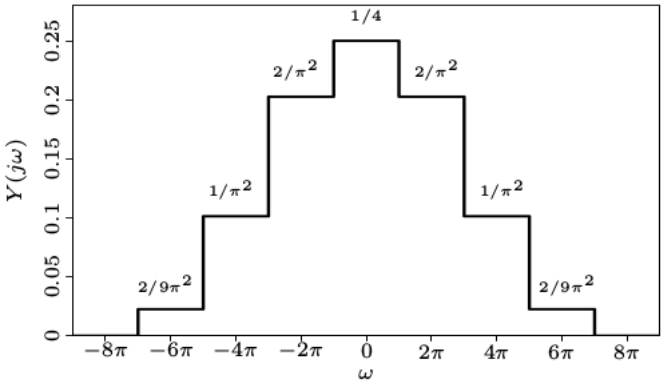


Figure 6: Sketch for $Y(j\omega)$ in Q6(c)

8. Given ,

$$\begin{aligned} z_1(t) &= x(t)\cos(\omega_1 t) + y(t)\cos(\omega_2 t) \\ \omega_1 &= 5W \\ \omega_2 &= 7W \end{aligned}$$

For $\mathbf{z_1(t)}$:

$$\begin{aligned} z_1(t) &= x(t)\cos(\omega_1 t) + y(t)\cos(\omega_2 t) \\ &= x(t)\Big(\frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2}\Big) + y(t)\Big(\frac{e^{j\omega_2 t} + e^{-j\omega_2 t}}{2}\Big) \end{aligned}$$

The frequency shift property of Fourier Transform states that,

$$f(t)e^{j\omega_0 t} \longleftrightarrow F(j(\omega - \omega_0))$$

Using this property and linearity, $Z_1(j\omega)$ can be written as,

$$\begin{aligned} Z_1(j\omega) = \frac{1}{2}\Big(&X(j(\omega-\omega_1)) + X(j(\omega+\omega_1)) + \\ &Y(j(\omega-\omega_2)) + Y(j(\omega+\omega_2))\Big) \end{aligned}$$

For $\mathbf{z_2(t)}$:

$Z_2(j\omega)$ is $Z_1(j\omega)$ bandlimited to $4W$ and $6W$

$$Z_2(j\omega) = \frac{1}{2}\Big(X(j(\omega-\omega_1)) + X(j(\omega+\omega_1))\Big)$$

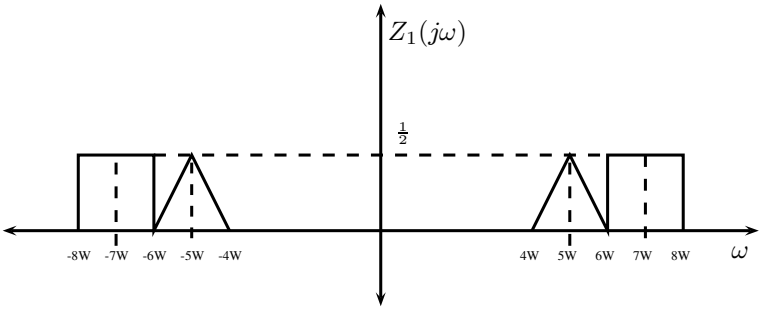


Figure 7: Sketch for $Z_1(j\omega)$

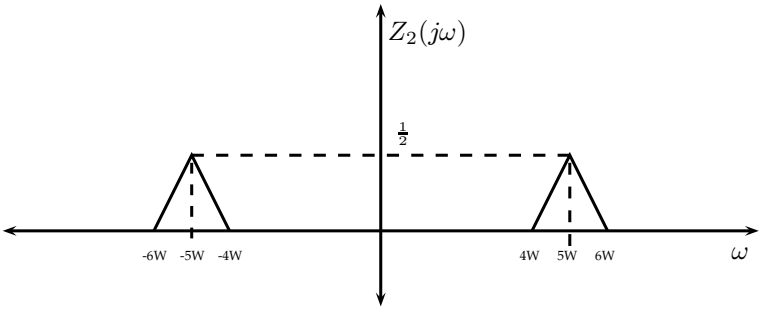


Figure 8: Sketch for $Z_2(j\omega)$

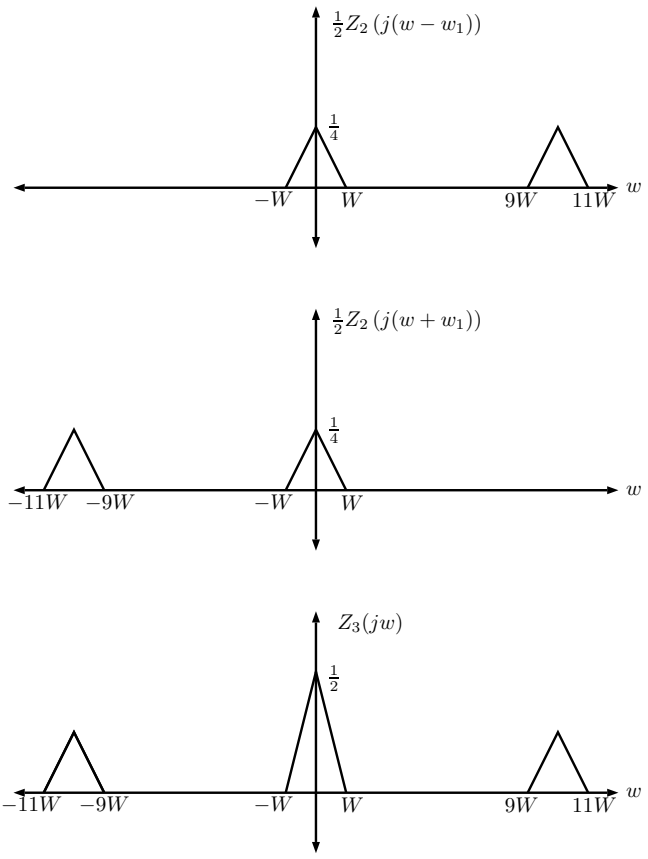


Figure 9: Sketch for $Z_3(j\omega)$

For $\mathbf{z_3(t)}$:

$$\begin{aligned} z_3(t) &= z_2(t)\cos(\omega t) \\ &= \frac{1}{2}\left(z_2(t)e^{j\omega_1 t} + z_2(t)e^{-j\omega_1 t}\right) \end{aligned}$$

Using frequency shifting property of Fourier Transform,

$$\begin{aligned} Z_3(j\omega) &= \frac{1}{2}\Big(Z_2(j(\omega-\omega_1)) + Z_2(j(\omega+\omega_1))\Big) \\ &= \frac{1}{4}\Big(X(j(\omega-2\omega_1)) + 2X(j\omega) + X(j(\omega+2\omega_1))\Big) \\ &= \frac{1}{2}X(j\omega) + \frac{1}{4}\Big(X(j(\omega-2\omega_1)) + X(j(\omega+2\omega_1))\Big) \end{aligned} \tag{2}$$

For $\mathbf{z_4(t)}$:

$z_4(t)$ is the LPF output for the input $z_3(t)$ with cutoff freq $= W$. So, only the first summation term from eqn.(2) is available as output in the form of $z_4(t)$, i.e.:

$$Z_4(j\omega) = \frac{1}{2}X(j\omega) \Rightarrow z_4(t) = \frac{1}{2}x(t)$$

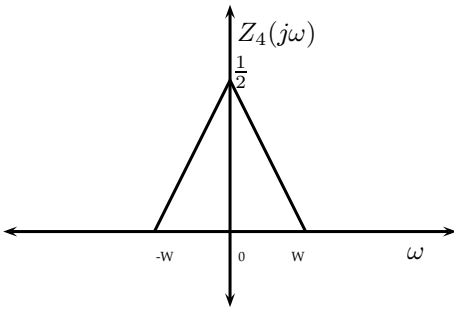


Figure 10: Sketch for $Z_4(j\omega)$